Computational reanalysis identifies intestinal and metabolic axes linking gut microbiota to Chronic Spontaneous Urticaria

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Abstract

Chronic spontaneous urticaria (CSU) is a persistent inflammatory skin disease characterized by immune dysregulation and altered metabolic signaling. Growing evidence suggests that **gut microbiota** contributes to CSU through immune modulation and production of bioactive metabolites. However, previous microbiome studies have reported inconsistent findings regarding which bacterial taxa and metabolic pathways are involved, mainly due to variations in sequencing pipelines, analytical tools, and reference databases. To address three discrepancies, we reanalyzed raw 16S rRNA gene sequencing data from three independent CSU cohorts using a **standardized pipeline**. We conducted differential abundance analyses with **MaAsLin2**, **Limma-voom**, and **Wilcoxon** tests, which helped identify microbial taxa that were consistently different between CSU and control groups. Functional potential was predicted with PICRUSt2, followed by contribution and correlation analyses linking taxa to KEGG orthologs (KOs). In all datasets, we found **82 taxa** with significant differences between groups. Among those frequently described in earlier reports, **Pseudomonadota**, **Enterobacteriaceae**, and ***Dialister*** were enriched in CSU, while ***Megamonas*** was consistently reduced. The functional analysis showed **212 KOs** that varied between groups, with 121 of these linked to the four taxa. Important features included K03399, which appeared in several taxa, and pathways related to **membrane transport**, **cofactor production**, **stress response**, and **DNA repair**. These functions reflect how the microbes adapt to inflammatory conditions. In contrast, functions associated with *Megamonas* suggested a **loss of metabolic balance** and a **decrease in anti-inflammatory capability**. This unified computational framework outlines two main microbial pathways-one that supports intestinal environmental adaptation and another that influences inflammatory metabolism. Together, these pathways offer a clear model linking microbial function to ongoing immune activation in CSU.

Background

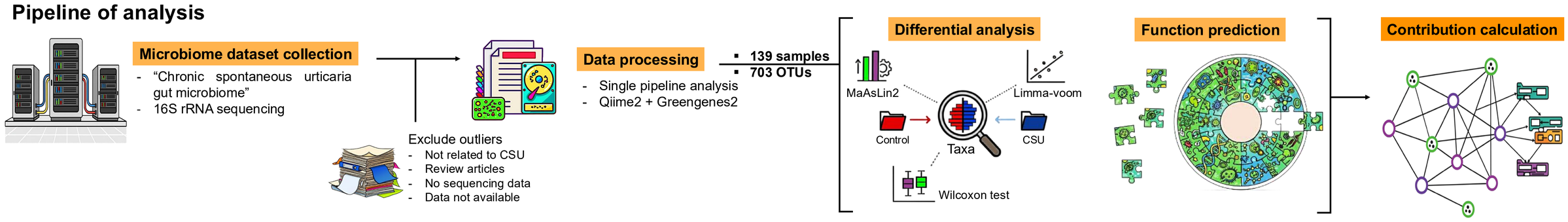
- Chronic spontaneous urticaria (CSU) is a recurrent inflammatory skin disorder with immune and metabolic dysregulation. Recent evidence suggests that gut microbiota influence CSU through metabolic and signaling interactions along the gut-skin-immune axis.
- Previous microbiome studies reported inconsistent taxa and functional patterns across cohorts. This lack of consensus is largely due to methodological challenges, such as differences in sequencing pipelines and reference databases, which limit reliable cross-study comparability.
- Functional mechanisms underlying microbial contributions to intestinal and systemic inflammation remain unclear. Therefore, this integrated study aims to reanalyze the datasets using a unified workflow to identify robust and reproducible microbial and functional signatures.

Objectives

To integrate multi-cohort microbiome data through a unified computational workflow and uncover intestinal and metabolic microbial pathways underlying CSU inflammation

Computational analysis

- **Pipeline of analysis**



- **Data standardization**

Data acquired from multiple public cohorts were processed through a unified Qiime2-Greengenes2 pipeline with consistent DADA2 denoising and taxonomy assignment, reducing inter-study variation and batch effects.

- **Taxonomic differential analysis**

Differential abundance was evaluated using MaAsLin2, Limma-voom, and Wilcoxon tests, and taxa significant in at least one method were considered reproducible candidates across datasets.

- **Functional integration**

Microbial functional profiles were predicted using PICRUSt2, and taxa-KO associations were examined to identify pathways linked to metabolism and microbial stress responses relevant to CSU.

- **Network correlation analysis**

Correlation networks between taxa and KEGG orthologs were constructed in Cytoscape to visualize functional connectivity and shared metabolic interactions among key taxa.

- **Study design and analytical framework for CSU meta-analysis**

Data source	PRJNA650123 [1]	PRJNA809140 [2]	PRJNA901136 [3]	Integrated meta-analysis
Study focus	Altered gut microbiota in chronic urticaria	Gut dysbiosis drives antihistamine-resistant chronic urticaria	Gut microbiota dysbiosis in chronic urticaria	Integration of public CSU 16S datasets to define core microbial and functional signatures
Sample size	20 CSU / 20 Control	25 CSU / 25 RCSU / 19 Control	15 CSU / 15 Control	85 CSU / 54 Control
Sequencing platform	Illumina HiSeq	Illumina MiSeq	Illumina MiSeq	Considered as confounder
Pipeline	Mothur + Greengenes2	Qiime2 + SILVA 138	Qiime2 + Greengenes2	Qiime2 + Greengenes2
Differential analysis	Metastats	LEfSe, Wilcoxon, PERMANOVA	LEfSe, Spearman	MaAsLin2 + Limma-voom + Wilcoxon

Reference papers
[1] Luo, Z., Jin, Z., Tao, X., Wang, T., Wei, P., Zhu, C., & Wang, Z. (2023). Combined microbiome and metabolome analysis of gut microbiota and metabolite interactions in chronic spontaneous urticaria. *Frontiers in Cellular and Infection Microbiology*, *12*, 1094737.
[2] Song, Y., Dan, K., Yao, Z., Yang, X., Chen, B., & Hao, F. (2022). Altered gut microbiota in H1-antihistamine-resistant chronic spontaneous urticaria associates with systemic inflammation. Frontiers in Cellular and Infection Microbiology, 12, 831489.
[3] Zhang, X., Zhang, J., Chu, Z., Shi, L., Geng, S., & Guo, K. (2021). Gut microbiome alterations and functional prediction in chronic spontaneous urticaria patients. Journal of Microbiology and Biotechnology, 31(5), 747.

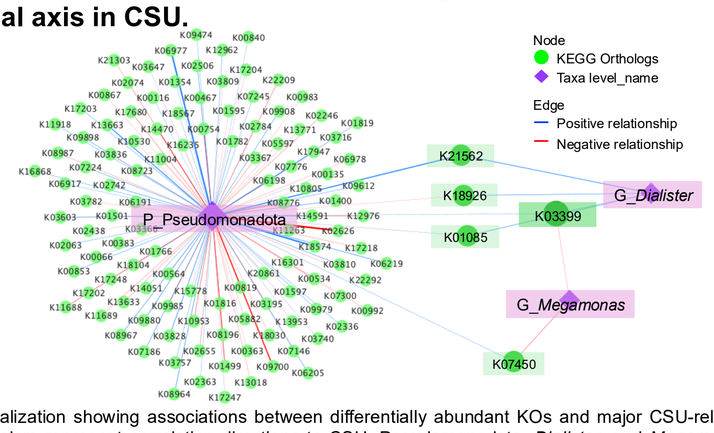
Results

Consistent enrichment of Pseudomonadota, Enterobacteriaceae, and *Dialister* and depletion of *Megamonas* in CSU across three reanalyzed datasets.

Taxa	Prior reports	Standarized analysis in integrated study
P_Pseudomonadota	Inconsistent	↑ MaAsLin2, Limma-voom, Wilcoxon
F_Enterobacteriaceae	Inconsistent	↑ MaAsLin2, Limma-voom, Wilcoxon
G_*Dialister*	Inconsistent	↑ MaAsLin2, Limma-voom, Wilcoxon
G_*Megamonas*	Decreased in CSU	↓ MaAsLin2, Limma-voom

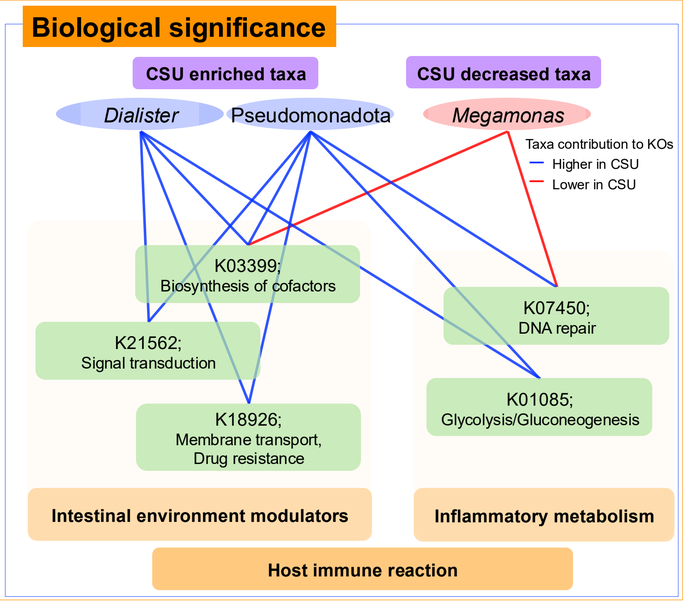
Summary of taxa identified as significantly different between CSU and control groups across three reanalyzed datasets. Analyses were performed using MaAsLin2, Limma-voom, and Wilcoxon tests under a unified pipeline. Arrows indicate the direction of change in CSU relative to controls. Pseudomonadota, Enterobacteriaceae, and *Dialister* were consistently enriched in CSU, whereas Megamonas showed a reproducible decrease across studies, partially resolving the conflicting results previously reported.

KO-taxa interaction network revealing a K03399-centered functional axis in CSU.



Network visualization showing associations between differentially abundant KOs and major CSU-related taxa. Edge colors represent correlation directions to CSU. Pseudomonadota, *Dialister*, and *Megamonas* were linked to 121 significant KOs inferred from PICRUSt2 contribution analysis. Among them, K03300 connected all three taxa, displaying positive associations with CSU-enriched taxa and a negative association with CSU-decreased *Megamonas*. Additional shared KOs also showed direction-specific relationships, indicating a potential microbial functional axis contributing to CSU pathophysiology.

Biological significance



Summary

- **Standardizing methodologies within a unified computational framework reveals robust disease-associated signatures**: Our study demonstrates that controlling analytical variability enables robust and reproducible microbiome analyses that reveal consistent biological signatures and clarify microbe–disease interactions. This integrative framework offers a transferable model for other microbiome-associated diseases and emphasizes the importance of reproducibility and multi-dimensional interpretation in identifying clinically actionable microbial signatures.
- **Our study uncovered a reproducible dysbiotic signature:** Pseudomonadota, Enterobacteriaceae, and *Dialister* were consistently enriched, while *Megamonas* was reduced in CSU, suggesting a microbial signature that may serve as a biomarker for the presence or severity of CSU.
- **Functional analysis reveals key pathways:** The enriched taxa are associated with pathways involving membrane transport, cofactor biosynthesis, stress response, and DNA repair, reflecting microbial adaptation to inflammatory conditions and their potential role in modulating host immune activation in CSU.

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